

31. Stacked Bar Plot with Error Bars

Aim

To write a Python program to create a stacked bar plot with error bars.

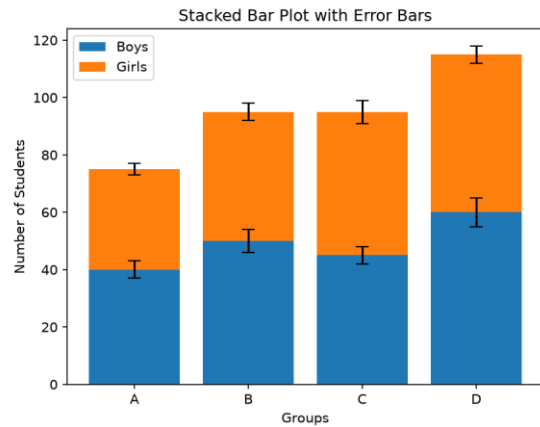
Dataset

Group	Boys	Girls	Boys Error	Girls Error
A	40	35	3	2
B	50	45	4	3
C	45	50	3	4
D	60	55	5	3

Python code:

```
1  import matplotlib.pyplot as plt
2  import numpy as np
3  groups = ["A", "B", "C", "D"]
4  boys = np.array([40, 50, 45, 60])
5  girls = np.array([35, 45, 50, 55])
6  boys_error = [3, 4, 3, 5]
7  girls_error = [2, 3, 4, 3]
8  plt.bar(groups, boys, yerr=boys_error, capsize=5, label="Boys")
9  plt.bar(groups, girls, bottom=boys,
10         yerr=girls_error, capsize=5, label="Girls")
11 plt.xlabel("Groups")
12 plt.ylabel("Number of Students")
13 plt.title("Stacked Bar Plot with Error Bars")
14 plt.legend()
15 plt.show()
```

Output:



Result:

A stacked bar plot with error bars was successfully created.

32. Scatter Graph Using Random Distribution

Aim

To write a Python program to draw a scatter graph using randomly distributed X and Y values.

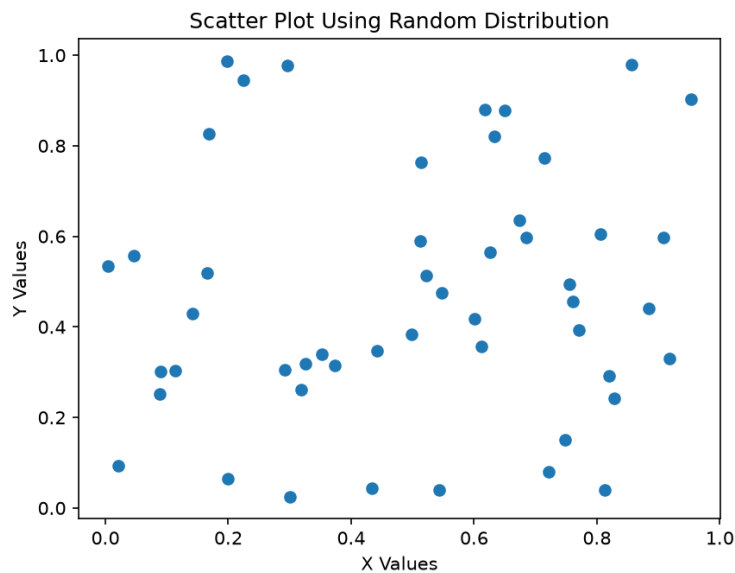
Dataset:

X	Y
Random	Random
Random	Random
Random	Random
Random	Random
Random	Random

Python

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3 np.random.seed(10)
4 x = np.random.rand(50)
5 y = np.random.rand(50)
6 plt.scatter(x, y)
7 plt.xlabel("X Values")
8 plt.ylabel("Y Values")
9 plt.title("Scatter Plot Using Random Distribution")
10 plt.show()
```

Output



Result

A scatter graph was successfully created using random X and Y distributions.